Chapter 12

# Respiration in Plants

All of us breathe to live, but why is breathing so essential to life? Do all living organisms, including plants and microbes, breathe, and if so, how?

## Energy for every activity

- all living organisms need energy for carrying out daily life activities, be it absorption, transport, movement, reproduction or even breathing
- the process of breathing is very much connected to the process of release of energy from food
- all the energy required for life processes is obtained by oxidation of some macromolecules that we call food

## Where the food comes from

- green plants and cyanobacteria can prepare their own food, trapping light energy by photosynthesis and converting it into chemical energy stored in the bonds of carbohydrates such as glucose, sucrose and starch
- even in green plants not all cells, tissues and organs photosynthesise, only cells containing chloroplasts, most often located in the superficial layers
- so all other organs, tissues and cells that are non-green need food for oxidation, and food has to be translocated to all non-green parts

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## 12.1 What respiration is

### Where the food ultimately comes from

- animals are heterotrophic, obtaining food from plants directly as herbivores or indirectly as carnivores, while saprophytes such as fungi depend on dead and decaying matter
- ultimately all the food that is respired for life processes comes from photosynthesis
- photosynthesis takes place within the chloroplasts in eukaryotes, whereas the breakdown of complex molecules to yield energy takes place in the cytoplasm and in the mitochondria

### The definition

- the breaking of the carbon to carbon bonds of complex compounds through oxidation within the cells, leading to release of a considerable amount of energy, is called respiration
- the compounds that are oxidised during this process are known as respiratory substrates
- usually carbohydrates are oxidised to release energy, but proteins, fats and even organic acids can be used as respiratory substances in some plants under certain conditions

## 12.2 Why ATP is the currency

### Energy released in steps

- during oxidation within a cell, all the energy contained in respiratory substrates is not released free into the cell, or in a single step
- it is released in a series of slow step-wise reactions controlled by enzymes, and is trapped as chemical energy in the form of ATP
- the energy released by oxidation in respiration cannot be used directly but is used to synthesise ATP, which is broken down whenever and wherever energy needs to be utilised, so ATP acts as the energy currency of the cell
- the carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in the cell

## 12.3 Do plants breathe?

### No special organs

- plants require $O_2$ for respiration to occur and they also give out $CO_2$, so they have systems in place that ensure the availability of $O_2$
- unlike animals they have no specialised organs for gaseous exchange, but they have stomata and lenticels for this purpose
- each plant part takes care of its own gas-exchange needs, and there is very little transport of gases from one plant part to another
- plants do not present great demands for gas exchange, roots, stems and leaves respiring at rates far lower than animals do

### Short distances, loose packing

- only during photosynthesis are large volumes of gases exchanged, and each leaf is well adapted to take care of its own needs during these periods; when cells photosynthesise, availability of O2 is not a problem since it is released within the cell
- the distance that gases must diffuse even in large, bulky plants is not great, each living cell being located quite close to the surface of the plant
- in stems the living cells are organised in thin layers inside and beneath the bark, and they have openings called lenticels, while the cells in the interior are dead and provide only mechanical support
- most cells of a plant have at least a part of their surface in contact with air, facilitated by the loose packing of parenchyma cells in leaves, stems and roots, which provides an interconnected network of air spaces

## 12.4 Why oxidation proceeds in small steps

### Combustion against respiration

- the complete combustion of glucose, which produces CO2 and H2O as end products, yields energy most of which is given out as heat, as $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy
- if this energy is to be useful to the cell, the cell should be able to utilise it to synthesise other molecules that it requires
- the strategy the plant cell uses is to catabolise the glucose molecule so that not all the liberated energy goes out as heat
- the key is to oxidise glucose not in one step but in several small steps, enabling some steps to be just large enough that the energy released can be coupled to ATP synthesis

### Life without oxygen

- during respiration oxygen is utilised, and carbon dioxide, water and energy are released as products, the combustion reaction requiring oxygen
- there are sufficient reasons to believe that the first cells on this planet lived in an atmosphere that lacked oxygen
- among present-day organisms several are adapted to anaerobic conditions, some being facultative anaerobes while in others the requirement for anaerobic condition is obligate
- in any case, all living organisms retain the enzymatic machinery to partially oxidise glucose without the help of oxygen, and this breakdown of glucose to pyruvic acid is called glycolysis

## 12.5 Glycolysis

### The EMP pathway

- the term comes from the Greek glycos for sugar and lysis for splitting
- the scheme was given by Gustav Embden, Otto Meyerhof and J. Parnas, and is often referred to as the EMP pathway; in anaerobic organisms it is the only process in respiration
- it occurs in the cytoplasm of the cell and is present in all living organisms, glucose undergoing partial oxidation to form two molecules of pyruvic acid

**Glucose**, phosphorylated to glucose-6-phosphate using ATP

**Fructose-6-phosphate**, then fructose-1,6-bisphosphate using a second ATP

**Split into two triose phosphates**

**Oxidised to bisphosphoglyceric acid**, reducing NAD+ to NADH

**Through phosphoglyceric acid and phosphoenolpyruvate to pyruvic acid**, yielding ATP

**Figure 12.A** Steps of glycolysis

### *Where the glucose comes from*

- in plants this glucose is derived from sucrose, the end product of photosynthesis, or from storage carbohydrates
- sucrose is converted into glucose and fructose by the enzyme invertase, and these two monosaccharides readily enter the glycolytic pathway
- glucose and fructose are phosphorylated to give glucose-6-phosphate by the activity of the enzyme hexokinase, which then isomerises to produce fructose-6-phosphate, after which the subsequent steps of metabolism of glucose and fructose are the same

## 12.6 Ten reactions from glucose to pyruvate

### *Where ATP is spent*

- ATP is utilised at two steps, first in the conversion of glucose into glucose-6-phosphate, and second in the conversion of fructose-6-phosphate to fructose-1,6-bisphosphate
- the fructose-1,6-bisphosphate is split into dihydroxyacetone phosphate and 3-phosphoglyceraldehyde, or PGAL

### *Where energy is gained*

- there is one step where NADH + H+ is formed from NAD+, when PGAL is converted to 1,3-bisphosphoglycerate, two redox equivalents being removed as two hydrogen atoms from PGAL and transferred to a molecule of NAD+
- PGAL is oxidised and, with inorganic phosphate, converted into BPGA
- the conversion of BPGA to 3-phosphoglyceric acid is also an energy yielding process, the energy being trapped by the formation of ATP
- another ATP is synthesised during the conversion of phosphoenolpyruvate to pyruvic acid, and pyruvic acid is then the key product of glycolysis

**Pyruvic acid at the branch point**

**Alcoholic fermentation**, giving CO2 and ethanol

**Lactic acid fermentation**, giving lactic acid

**NADH reoxidised to NAD+ in both routes**

**Figure 12.B** Major pathways of anaerobic respiration

### Three fates for pyruvate

- there are three major ways in which different cells handle pyruvic acid produced by glycolysis, namely lactic acid fermentation, alcoholic fermentation and aerobic respiration
- fermentation takes place under anaerobic conditions in many prokaryotes and unicellular eukaryotes
- for the complete oxidation of glucose to CO2 and H2O, however, organisms adopt the Krebs cycle, also called aerobic respiration, which requires an O2 supply

## 12.7 Fermentation

### Two routes

- in fermentation, say by yeast, the incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO2 and ethanol
- the enzymes pyruvic acid decarboxylase and alcohol dehydrogenase catalyse these reactions
- other organisms such as some bacteria produce lactic acid from pyruvic acid, and in animal cells too, as in muscles during exercise when oxygen is inadequate, pyruvic acid is reduced to lactic acid by lactate dehydrogenase
- the reducing agent is NADH + H+, which is reoxidised to NAD+ in both the processes

### Why the yield is poor

- in both lactic acid and alcohol fermentation not much energy is released, less than seven per cent of the energy in glucose, and not all of it is trapped as high energy bonds of ATP
- the processes are also hazardous, since either acid or alcohol is produced
- yeasts poison themselves to death when the concentration of alcohol reaches about 13 per cent

## 12.8 Aerobic respiration

### The complete oxidation

- aerobic respiration leads to a complete oxidation of organic substances in the presence of oxygen, releasing CO2, water and a large amount of the energy present in the substrate, and is most common in higher organisms
- for it to take place within the mitochondria, the final product of glycolysis, pyruvate, is transported from the cytoplasm into the mitochondria
- the crucial events are the complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO2
- and the passing on of the electrons removed as part of the hydrogen atoms to molecular O2, with simultaneous synthesis of ATP

### Where each event happens

- the first process takes place in the matrix of the mitochondria
- the second is located on the inner membrane of the mitochondria

### Oxidative decarboxylation

- pyruvate, formed by the glycolytic catabolism of carbohydrates in the cytosol, after entering the mitochondrial matrix undergoes oxidative decarboxylation by a complex set of reactions catalysed by pyruvic dehydrogenase
- these reactions require the participation of several coenzymes, including NAD+ and Coenzyme A
- the reaction is pyruvic acid + CoA + NAD+ → acetyl CoA + CO2 + NADH + H+
- during this process two molecules of NADH are produced from the metabolism of two molecules of pyruvic acid, produced from one glucose molecule during glycolysis

## 12.9 The tricarboxylic acid cycle

### How the cycle starts

- the acetyl CoA enters a cyclic pathway, the tricarboxylic acid cycle, more commonly called the Krebs cycle after the scientist Hans Krebs who first elucidated it
- the cycle starts with the condensation of the acetyl group with oxaloacetic acid and water to yield citric acid, catalysed by the enzyme citrate synthase, with a molecule of CoA released
- citrate is then isomerised to isocitrate, followed by two successive steps of decarboxylation, leading to the formation of alpha-ketoglutaric acid and then succinyl-CoA

**Pyruvate entering as acetyl coenzyme A**, releasing $CO_2$

**Citric acid formed from acetyl CoA and oxaloacetic acid**

**Through alpha-ketoglutaric acid and succinic acid**, releasing two more $CO_2$

**Malic acid returning to oxaloacetic acid to continue the cycle**

**NADH**, $FADH_2$ and GTP produced along the way

**Figure 12.C** The Citric acid cycle

### The remaining steps

- succinyl-CoA is oxidised to oxaloacetic acid, allowing the cycle to continue
- during the conversion of succinyl-CoA to succinic acid a molecule of GTP is synthesised, which is a substrate level phosphorylation, and in a coupled reaction GTP is converted to GDP with the simultaneous synthesis of ATP from ADP
- there are three points in the cycle where $NAD^+$ is reduced to $NADH + H^+$ and one point where $FAD^+$ is reduced to $FADH_2$
- the continued oxidation of acetyl CoA requires the continued replenishment of oxaloacetic acid, the first member of the cycle, and regeneration of $NAD^+$ and $FAD^+$ from NADH and $FADH_2$

### The balance so far

- glucose has been broken down to release $CO_2$, and eight molecules of $NADH + H^+$ and two of $FADH_2$ have been synthesised, besides just two molecules of ATP in the TCA cycle
- neither $O_2$ has come into the picture nor has the large number of ATP yet been synthesised

## 12.10 The electron transport system

### Releasing the stored energy

- the following steps release and utilise the energy stored in $NADH + H^+$ and $FADH_2$
- this is accomplished when they are oxidised through the electron transport system and the electrons are passed on to $O_2$, resulting in the formation of $H_2O$
- the metabolic pathway through which the electron passes from one carrier to another is called the electron transport system, and it is present in the inner mitochondrial membrane

**Complex I**, NADH dehydrogenase

**Complex II**, feeding in FADH2 from succinate

**Ubiquinone**, then complex III, the cytochrome bc1 complex

**Cytochrome c as mobile carrier**, then complex IV, cytochrome c oxidase

**Complex V**, ATP synthase, making ATP from ADP

**Figure 12.D** Electron Transport System (ETS)

### The carriers in order

- electrons from NADH produced in the mitochondrial matrix during the citric acid cycle are oxidised by an NADH dehydrogenase, complex I, and the electrons are then transferred to ubiquinone located within the inner membrane
- ubiquinone also receives reducing equivalents via FADH2, complex II, generated during oxidation of succinate in the citric acid cycle
- the reduced ubiquinone, ubiquinol, is then oxidised with the transfer of electrons to cytochrome c via the cytochrome bc1 complex, complex III
- cytochrome c is a small protein attached to the outer surface of the inner membrane and acts as a mobile carrier between complex III and complex IV, the cytochrome c oxidase complex containing cytochromes a and a3 and two copper centres

### How much ATP

- when the electrons pass from one carrier to another via complexes I to IV they are coupled to ATP synthase, complex V, for the production of ATP from ADP and inorganic phosphate
- the number of ATP molecules synthesised depends on the nature of the electron donor
- oxidation of one molecule of NADH gives rise to 3 molecules of ATP, while that of one molecule of FADH2 produces 2 molecules of ATP

### The role of oxygen

- although the aerobic process takes place only in the presence of oxygen, the role of oxygen is limited to the terminal stage of the process
- yet the presence of oxygen is vital, since it drives the whole process by removing hydrogen from the system, acting as the final hydrogen acceptor
- unlike photophosphorylation, where light energy is utilised for the production of the proton gradient, in respiration it is the energy of oxidation-reduction that is utilised, and for this reason the process is called oxidative phosphorylation

**F1 headpiece**, a peripheral membrane protein complex

**It carries the site for synthesis of ATP from ADP and inorganic phosphate**

**F0**, an integral membrane protein complex

**It forms the channel through which protons cross the inner membrane**

**Figure 12.E** Diagramatic presentation of ATP synthesis in mitochondria

## 12.11 ATP synthase

### Two components

- the energy released during the electron transport system is utilised in synthesising ATP with the help of ATP synthase, complex V, which consists of two major components, F1 and F0
- the passage of protons through the channel is coupled to the catalytic site of the F1 component for the production of ATP
- for each ATP produced, 4H+ pass through F0 from the intermembrane space to the matrix, down the electrochemical proton gradient

## 12.12 The respiratory balance sheet

### The assumptions behind the count

- there is a sequential, orderly pathway functioning, with one substrate forming the next, and with glycolysis, TCA cycle and ETS following one after another
- the NADH synthesised in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation
- none of the intermediates in the pathway are utilised to synthesise any other compound
- only glucose is being respired, no other alternative substrates entering at any of the intermediary stages

### Why they do not hold

- in a living system all pathways work simultaneously and do not take place one after another
- substrates enter the pathways and are withdrawn from them as and when necessary, ATP is utilised as and when needed, and enzymatic rates are controlled by multiple means
- yet it is useful to do this exercise to appreciate the beauty and efficiency of the living system in extracting and storing energy
- there can be a net gain of 38 ATP molecules during aerobic respiration of one molecule of glucose

## 12.13 Fermentation against aerobic respiration

### Three differences

- fermentation accounts for only a partial breakdown of glucose, whereas in aerobic respiration it is completely degraded to $CO_2$ and $H_2O$
- in fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid, whereas many more are generated under aerobic conditions
- NADH is oxidised to NAD+ rather slowly in fermentation, however the reaction is very vigorous in aerobic respiration

## 12.14 The amphibolic pathway

### Where other substrates enter

- glucose is the favoured substrate for respiration, and all carbohydrates are usually first converted into glucose before being used
- other substrates can also be respired, but they do not enter the respiratory pathway at the first step
- fats would need to be broken down into glycerol and fatty acids first, fatty acids being degraded to acetyl CoA before entering, while glycerol enters after being converted to PGAL
- proteins would be degraded by proteases, and the individual amino acids after deamination would enter at some stage within the Krebs cycle, or even as pyruvate or acetyl CoA

### Both breakdown and synthesis

- since respiration involves breakdown of substrates, the respiratory process has traditionally been considered a catabolic process
- but the very compounds that enter would also be withdrawn from the respiratory pathway for the synthesis of those same substrates
- fatty acids are broken down to acetyl CoA before entering when used as a substrate, but when the organism needs to synthesise fatty acids, acetyl CoA is withdrawn from the pathway for it
- breaking down within the living organism is catabolism and synthesis is anabolism, and since the respiratory pathway is involved in both, it is better considered an amphibolic pathway rather than a catabolic one

**Carbohydrates**, fats and proteins entering at different points

**Fatty acids entering as acetyl CoA**, glycerol as PGAL

**Amino acids entering at the Krebs cycle**, as pyruvate or as acetyl CoA

**All routes leading to CO2 and water**

**Figure 12.F** Interrelationship among metabolic pathways showing respiration mediated breakdown of different organic molecules to CO2 and H20

## 12.15 Respiratory quotient

### The ratio

- during aerobic respiration O2 is consumed and CO2 is released
- the ratio of the volume of CO2 evolved to the volume of O2 consumed in respiration is called the respiratory quotient, or respiratory ratio
- the respiratory quotient depends upon the type of respiratory substrate used during respiration

### Carbohydrates

- when carbohydrates are used as substrate and are completely oxidised, the quotient will be 1
- this is because equal amounts of CO2 and O2 are evolved and consumed respectively

### The three substrates

- for carbohydrates, $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy, so the quotient of $6CO_2$ to $6O_2$ is 1.0
- when fats are used the quotient is less than 1, and for the fatty acid tripalmitin, $2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98H_2O$ + energy, giving a quotient of 102 to 145, that is 0.7
- when proteins are respiratory substrates the ratio would be about 0.9
- in living organisms respiratory substrates are often more than one, and pure proteins or fats are never used as respiratory substrates

## 12.16 Summary

### Gaseous exchange and glycolysis

- plants, unlike animals, have no special systems for breathing or gaseous exchange, stomata and lenticels allowing gaseous exchange by diffusion, and almost all living cells in a plant have their surfaces exposed to air
- the breaking of carbon to carbon bonds of complex organic molecules by oxidation, leading to the release of a lot of energy, is called cellular respiration, glucose being the favoured substrate, though fats and proteins can also be broken down to yield energy
- the initial stage takes place in the cytoplasm, each glucose molecule being broken through a series of enzyme catalysed reactions into two molecules of pyruvic acid, a process called glycolysis
- the fate of the pyruvate depends on the availability of oxygen and the organism, and under anaerobic conditions either lactic acid fermentation or alcohol fermentation occurs

### Aerobic respiration

- fermentation takes place under anaerobic conditions in many prokaryotes, unicellular eukaryotes and in germinating seeds
- in eukaryotic organisms aerobic respiration occurs in the presence of oxygen, pyruvic acid being transported into the mitochondria where it is converted into acetyl CoA with the release of $CO_2$
- acetyl CoA then enters the tricarboxylic acid pathway or Krebs cycle operating in the matrix of the mitochondria, where NADH + H+ and $FADH_2$ are generated
- the energy in these molecules, as well as that in the NADH + H+ synthesised during glycolysis, is used to synthesise ATP

### The electron transport system, and the pathway as a whole

- this is accomplished through a system of electron carriers called the electron transport system, located on the inner membrane of the mitochondria
- the electrons, as they move through the system, release enough energy that is trapped to synthesise ATP, and this is called oxidative phosphorylation, in which O2 is the ultimate acceptor of electrons and gets reduced to water
- the respiratory pathway is an amphibolic pathway, as it involves both anabolism and catabolism
- the respiratory quotient depends upon the type of respiratory substance used during respiration

## 12.17 Exercises

### From this chapter

1. Differentiate between (a) respiration and combustion (b) glycolysis and the Krebs cycle (c) aerobic respiration and fermentation
2. What are respiratory substrates? Name the most common respiratory substrate.
3. Give the schematic representation of glycolysis.
4. What are the main steps in aerobic respiration? Where does it take place?
5. Give the schematic representation of an overall view of the Krebs cycle.
6. Explain the electron transport system.
7. Distinguish between the following: (a) aerobic respiration and anaerobic respiration (b) glycolysis and fermentation (c) glycolysis and the citric acid cycle
8. What are the assumptions made during the calculation of net gain of ATP?
9. Discuss the statement that the respiratory pathway is an amphibolic pathway.
10. Define the respiratory quotient. What is its value for fats?
11. What is oxidative phosphorylation?
12. What is the significance of step-wise release of energy in respiration?